Logic gates

Computers carry out calculations on data represented by binary numbers. These calculations are done at the lowest level of the computer's hardware using devices called logic gates.

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Making decisions

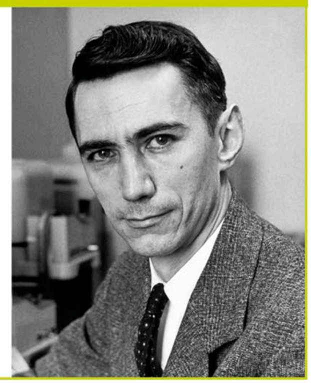
Logic gates are the building blocks of digital computers as they help in making decisions. They are electronic components whose output depends on their input, following the rules of Boolean logic – a form of algebra where values are either TRUE or FALSE. All possible values of input to a logic gate and the corresponding output can be shown in a truth table. In a truth table, the binary value 1 is equal to the logical value TRUE and the binary value 0 is equal to the logical value FALSE.

▽ Logic gates, truth tables, and circuits
The table below shows the seven logic gates and their corresponding truth tables. Logic gates can also be combined to make circuits. Constructing a truth table for a circuit helps us to predict how it will behave.

BIOGRAPHY

Claude Shannon

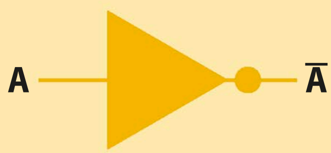
American mathematician Claude Shannon (1916–2001) made real-world versions of Boolean logic by using electrical switches, with ON and OFF representing the values TRUE and FALSE. Shannon then developed combinations of electrical switches capable of making decisions or calculating numerical values – forming the basis of modern digital computing.



Logic gate	Symbol	Truth table

NOT

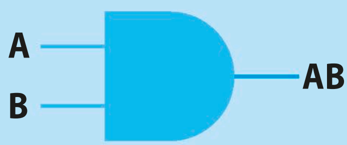
A NOT gate's output is the opposite of its input. If the input is 0, its output is 1, and if its input is 1, the output is 0.



INPUT	OUTPUT
0	1
1	0

AND

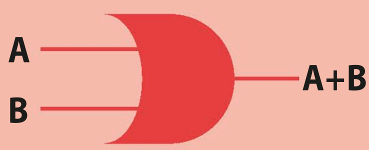
An AND gate's output is 1 only if both its inputs are 1.



INPUT		OUTPUT
A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

OR

An OR gate's output is 1 if either or both its inputs are 1. This is sometimes known as Inclusive-OR.



INPUT		OUTPUT
A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1